Light-Induced Topological Phase Transitions and Anomalous Thermal Transport in d-Wave Altermagnets

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We study intrinsic thermal transport and Floquet-engineered topology in a two-dimensional d-wave altermagnetic topological insulator powered by linearly polarized light. We analyze the anomalous Hall, Nernst, and thermal Hall conductivities, as well as their spin-resolved equivalents, and develop closed-form formulas for the Berry curvature using an analytically calculated high-frequency effective Hamiltonian. We demonstrate that linearly polarized light, in contrast to conventional antiferromagnets, breaks the symmetry connecting spin sectors in altermagnets, allowing a series of spin-selective topological phase transitions from a quantum spin Hall state to a spin-polarized Chern insulator and finally to a trivial phase. The Nernst response shows substantial thermal activation and significant sensitivity to the gap size in the Chern domain, but both the electrical and thermal Hall responses become quantized and meet the anomalous Wiedemann–Franz law. Every anomalous transport coefficient exhibits a distinctive d-wave dependence on the polarization angle, reversing sign under orthogonal rotation and vanishing at symmetry-restoring directions. Our findings show a path to all-optical regulation of topological and caloritronic responses beyond traditional magnetic systems and establish thermal transport as a sensitive probe of altermagnetic order.

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I. INTRODUCTION

Altermagnetism (AM) has recently emerged as a distinct magnetic phase characterized by zero net magnetization coexisting with strongly anisotropic, momentum-dependent spin splitting [1–3], and has been identified in a growing number of materials [4–8]. Unlike conventional antiferromagnets (AFMs), which preserve combined parity–time ($\mathcal{PT}$) symmetry and exhibit spin-degenerate bands, altermagnets break both time-reversal ($\mathcal{T}$) and $\mathcal{PT}$ symmetries while preserving their combination with specific crystal rotations [9]. This distinctive symmetry structure [10] enables a wide range of spintronic and topological phenomena, including giant magnetoresistance [3], anomalous Hall effects [11], topological insulating phases [12–14], and unconventional superconducting proximity effects [15].

Light is a particularly powerful external stimulus for controlling quantum materials, as it couples directly to electronic degrees of freedom and can dynamically modify band structure and symmetry. Beyond enabling control of intrinsic electronic and magnetic properties, time-periodic driving can induce nonequilibrium phases that have no static counterparts. Within the framework of Floquet theory [16–18], such periodic driving provides a versatile route to engineer effective Hamiltonians and realize dynamically generated topological states [19–21].

In conventional nonmagnetic and antiferromagnetic systems, Floquet engineering has enabled a wide range of phenomena, including light-induced Hall effects [22–24], Floquet topological insulators [16, 25], light-induced Weyl semimetals [26], and Floquet time crystals [27]. It has also opened a route to ultrafast control of spin and magnetic textures on picosecond and sub-picosecond timescales, forming the basis of ultrafast spintronics [18, 28]. In topological systems, light-driven Floquet states have enabled novel phenomena such as Majorana modes, light-controlled edge transport, and dynamical probes of topological phase transitions.

Recently, the effects of light on unconventional magnetic systems, particularly altermagnets, have begun to attract significant attention [9, 29–31]. However, existing studies have largely focused on charge transport and high-frequency regimes. While Berry-curvature-driven thermal transport is well established in topological materials [32], its Floquet engineering in altermagnetic systems remains essentially unexplored. This is especially important because symmetry considerations allow linearly polarized light (LPL) to induce Hall responses in altermagnets, in contrast to conventional antiferromagnets.

In nonmagnetic and conventional antiferromagnetic systems, LPL generally cannot generate a Hall response because $\mathcal{PT}$ symmetry is preserved. Altermagnets circumvent this restriction: their intrinsically broken $\mathcal{PT}$ symmetry allows LPL, by breaking the spin-rotation symmetry $C_{4z}\mathcal{T}$ that relates the two spin sublattices, to induce a finite anomalous Hall effect [9]. Recent works have demonstrated LPL-induced anomalous Hall responses and topological phase transitions in altermagnetic topological insulators [9], as well as circularly polarized light–induced quantum anomalous Hall phases [29], light-induced odd-parity magnetism [30, 33], and higher-order spin–orbit coupling effects [31].

Despite this progress, a comprehensive understanding of thermal transport under LPL irradiation in altermag-

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netics remains lacking. Thermal transport coefficients; namely the anomalous Nernst conductivity (ANC) and anomalous thermal Hall conductivity (ATHC), probe the Berry curvature through distinct thermal weightings that differ fundamentally from the anomalous Hall conductivity (AHC). As a result, they provide complementary and often more sensitive insights into the underlying topology and magnetic order. The ANC, being thermally activated, serves as a sensitive probe of band topology at finite temperature, while the ATHC, together with its relation to σ_{xy} via the Wiedemann–Franz (WF) law, offers a direct test of topological quantization. Furthermore, the spin-resolved counterparts; the spin Nernst and spin thermal Hall conductivities, enable access to spin caloritronic effects of direct relevance for device applications [25].

In this work, we present a comprehensive theoretical study of thermal transport in a d-wave altermagnetic under LPL irradiation. Starting from an analytically derived Floquet effective Hamiltonian in the high-frequency limit, we obtain explicit expressions for the Berry curvature in each spin sector and compute six intrinsic transport coefficients: the AHC σ_{xy}, ANC α_{xy}, ATHC $\kappa_{xy}^{(0)}$, as well as their spin-resolved counterparts σ_{xy}^{s}, α_{xy}^{s}, and $\kappa_{xy}^{(0),s}$. We analyze their dependence on topological phase, polarization angle, and temperature, and establish their interrelations via the Mott and Wiedemann–Franz laws. Finally, we contrast the altermagnetic response with that of conventional antiferromagnets, identifying clear experimental signatures accessible through thermal Hall and Nernst measurements.

II. MODEL AND FLOQUET EFFECTIVE HAMILTONIAN

We employ the four-band two-dimensional out-of-plane d-wave altermagnetic model on a square lattice [9, 34, 35] in which the orbitals are $p_{z\uparrow}$, $\frac{1}{\sqrt{2}}(d_{xz\uparrow} + id_{yz\uparrow})$, $p_{z\downarrow}$, and $\frac{1}{\sqrt{2}}(d_{xz\downarrow} - id_{yz\downarrow})$. The low-energy Hamiltonian has the block-diagonal form

$$H(\boldsymbol{k}) = \begin{pmatrix} H_{\uparrow}(\boldsymbol{k}) & 0 \\ 0 & H_{\downarrow}(\boldsymbol{k}) \end{pmatrix}, \tag{1}$$

where each 2×2 spin block is written in terms of a d-vector as $H_{\sigma}(\boldsymbol{k}) = \boldsymbol{d}^{\sigma}(\boldsymbol{k}) \cdot \boldsymbol{\tau}$, with $\boldsymbol{\tau} = (\tau_x, \tau_y, \tau_z)$ the Pauli matrices acting on the orbital subspace. The explicit d-vector components are

$$\begin{aligned} d_x^{\uparrow} &= -v \sin k_x, \quad d_y^{\uparrow} = -vt_a \sin k_y, \\ d_z^{\uparrow} &= m + b(\cos k_x + t_a^2 \cos k_y), \end{aligned} \tag{2}$$

and $H_{\downarrow}(k_x, k_y) = H_{\uparrow}(k_y, k_x)$, i.e., the spin-down block is obtained by the exchange $k_x \leftrightarrow k_y$. Here m is the on-site potential, v the hopping amplitude, b the inversion-breaking hopping, and t_a the asymmetry parameter. The two spin channels are related by $C_{4z}\mathcal{T}$ or mirror symmetry $\mathcal{M}_{xy}$. The lattice constant is assumed to be unity. Setting $t_a = 1$ restores $\mathcal{PT}$ symmetry, reducing the system to a conventional AFM; $t_a \neq 1$ breaks $\mathcal{PT}$ and yields the d-wave AM. The equilibrium system is a QSH insulator when $|m| < |b|(1 + t_a^2)$ and topologically trivial otherwise [12].

We drive the system with LPL whose vector potential is given as

$$\boldsymbol{A}(t) = A_0[\cos\theta \cos(\omega t), \sin\theta \cos(\omega t)], \tag{3}$$

where A_0 is the amplitude (it is dimensionless in our units), θ the polarization angle relative to the x-axis, and ω the frequency. The time-dependent Hamiltonian is obtained via the Peierls substitution $\boldsymbol{k} \to \boldsymbol{k} + e\boldsymbol{A}(t)/\hbar$. In the high frequency limit and off-resonant regime, the leading order commutator correction vanishes because $\boldsymbol{A}(t) = \boldsymbol{A}(-t + \tau)$ for a fractional time translation τ, and the effective Hamiltonian reduces to the time averaged part [19, 20] in Floquet high-frequency expansion regime. In the off-resonant limit, the heating rate is significantly reduced, allowing the system to reach a long-lived Floquet prethermal [36, 37] state where the effective Hamiltonian description remains valid or the situation where a finite pulse duration in order of ps or ns is used. Finite temperature primarily broadens the Fermi distribution, however, does not destroy the topological band structure provided $k_B T \ll \hbar\omega$. We assume that higher-order Floquet corrections do not qualitatively modify the topological phases in the studied system. Applying the Jacobi Anger expansion gives renormalized d-vectors:

$$\begin{aligned} d_{x,\text{eff}}^{\uparrow} &= -vj_1 \sin k_x, \quad d_{y,\text{eff}}^{\uparrow} = -vt_a j_2 \sin k_y, \\ d_{z,\text{eff}}^{\uparrow} &= m + b[j_1 \cos k_x + j_2 t_a^2 \cos k_y], \end{aligned} \tag{4}$$

$$\begin{aligned} d_{x,\text{eff}}^{\downarrow} &= -vj_2 \sin k_y, \quad d_{y,\text{eff}}^{\downarrow} = -vt_a j_1 \sin k_x, \\ d_{z,\text{eff}}^{\downarrow} &= m + b[j_2 \cos k_y + j_1 t_a^2 \cos k_x], \end{aligned} \tag{5}$$

where

$$j_1 = J_0(A_0 \cos\theta), \quad j_2 = J_0(A_0 \sin\theta), \tag{6}$$

and J_0 is the zeroth-order Bessel function. The key physical content of Eqs. (4)–(6) is transparent: when $\theta \neq \pm\pi/4$, we have $j_1 \neq j_2$, which means the spin-up and spin-down effective Hamiltonians are no longer related by any symmetry operation, breaking $C_{4z}\mathcal{T}$ and $\mathcal{M}_{xy}$. In contrast, for the conventional AFM ($t_a = 1$), the $\mathcal{PT}$ symmetry is preserved for any θ, locking $j_1 = j_2$ in its effect on the combined system and hinders any Hall response.

The eigenvalues of H_{eff}^{σ} are $\varepsilon_{\pm}^{\sigma}(\boldsymbol{k}) = \pm|\boldsymbol{d}_{\text{eff}}^{\sigma}(\boldsymbol{k})|$. A topological phase transition occurs whenever a spin-resolved gap $\Delta_{\sigma} = \min_{\boldsymbol{k}}[2|\boldsymbol{d}_{\text{eff}}^{\sigma}(\boldsymbol{k})|]$ closes. Since $j_1 \neq j_2$ under LPL in the AM, the two gaps $\Delta_{\uparrow}$ and $\Delta_{\downarrow}$ evolve independently with A_0, enabling sequential closings that are forbidden in the AFM.

III. BERRY CURVATURE

The spin-up and spin- down components in the Eq. (1) are decoupled, their Berry connections and Chern numbers can be evaluated independently. For the lower band $\varepsilon_-^\sigma = -|\boldsymbol{d}^\sigma|$ of each spin block, the normalized eigenstate in terms of the spherical coordinates $\hat{d}^\sigma = (\sin\Theta\cos\Phi, \sin\Theta\sin\Phi, \cos\Theta)$ of the effective d-vector is

$$|u_-^\sigma(\boldsymbol{k})\rangle = \begin{pmatrix} -\sin(\Theta/2)\, e^{-i\Phi} \\ \cos(\Theta/2) \end{pmatrix}. \tag{7}$$

Computing the Berry connection $\mathcal{A}_\mu^\sigma = i\langle u_-^\sigma|\partial_{k_\mu}|u_-^\sigma\rangle$ from Eq. (7) gives

$$\mathcal{A}_\mu^\sigma(\boldsymbol{k}) = \sin^2\left(\frac{\Theta}{2}\right)\partial_{k_\mu}\Phi. \tag{8}$$

Taking the antisymmetric curl of Eq. (8) and converting to the solid angle form, the z-component of the Berry curvature for the lower band of spin σ can be expressed as

$$\Omega_-^{z,\sigma}(\boldsymbol{k}) = \frac{1}{2}\hat{d}^\sigma\cdot\left(\partial_{k_x}\hat{d}^\sigma \times \partial_{k_y}\hat{d}^\sigma\right) = \frac{\boldsymbol{d}^\sigma\cdot(\partial_{k_x}\boldsymbol{d}^\sigma \times \partial_{k_y}\boldsymbol{d}^\sigma)}{2|\boldsymbol{d}^\sigma|^3}. \tag{9}$$

Evaluating the triple product explicitly with the LPL renormalized d-vectors (4) and (5), we obtain the closed form expression for the spin-up Berry curvature as:

$$\Omega_-^{z,\uparrow}(\boldsymbol{k}) = \frac{bv^2t_aj_1^2j_2\sin^2 k_x\cos k_y + bv^2t_a^3j_1j_2^2\sin^2 k_y\cos k_x}{2|\boldsymbol{d}_{\rm eff}^\uparrow|^3} + \frac{d_{z,\rm eff}^\uparrow\, v^2t_aj_1j_2\cos k_x\cos k_y}{2|\boldsymbol{d}_{\rm eff}^\uparrow|^3}, \tag{10}$$

and for spin-down:

$$\Omega_-^{z,\downarrow}(\boldsymbol{k}) = \frac{-bv^2t_aj_1j_2^2\sin^2 k_y\cos k_x - bv^2t_a^3j_1^2j_2\sin^2 k_x\cos k_y}{2|\boldsymbol{d}_{\rm eff}^\downarrow|^3} - \frac{d_{z,\rm eff}^\downarrow\, v^2t_aj_1j_2\cos k_x\cos k_y}{2|\boldsymbol{d}_{\rm eff}^\downarrow|^3}. \tag{11}$$

Equations (10) and (11) encode several important symmetry properties. First, when $j_1 = j_2$ (i.e., $A_0 = 0$ or $\theta = \pm\pi/4$), we can verify directly that $\Omega_-^{z,\uparrow}(\boldsymbol{k}) = -\Omega_-^{z,\downarrow}(\boldsymbol{k})$, so the total Berry curvature $\Omega_-^{z,\uparrow} + \Omega_-^{z,\downarrow} = 0$ at every $\boldsymbol{k}$, consistent with the preserved $C_{4z}\mathcal{T}$ symmetry. Second, under $\mathcal{PT}$ symmetry (AFM, $t_a = 1$, any j_1, j_2): $\mathcal{PT}$ maps $\Omega_n^z(\boldsymbol{k}) \to -\Omega_n^z(\boldsymbol{k})$, forcing $\Omega_n^z(\boldsymbol{k}) = 0$ for all $\boldsymbol{k}$, which prevents any Hall response. Third, the symmetry $\sigma_{xy}(\pi/2-\theta) = -\sigma_{xy}(\theta)$ is encoded in Eqs. (10)–(11) through the exchange $j_1 \leftrightarrow j_2$ under $\theta \to \pi/2-\theta$.

IV. THERMAL TRANSPORT

Having calculated the Berry phase, we can calculate some thermal transport quantities like anomalous Hall conductivity, Nernst conductivity and thermal Hall conductivity.

A. Anomalous Hall conductivity

The low-energy and effective Hamiltonian is block diagonal in spin space thus the spin S_z is a good quantum number, the valence bands split into spin-up and spin-down components. Therefore, we can define the spin current in the system. Within the semiclassical Boltzmann framework, the anomalous (intrinsic) contribution to the transverse conductivity is [38]

$$\sigma_{xy} = \frac{e^2}{\hbar}\sum_\sigma\int\frac{d^2k}{(2\pi)^2}\,\Omega_-^{z,\sigma}(\boldsymbol{k})\, f(\varepsilon_-^\sigma(\boldsymbol{k})-\mu), \tag{12}$$

where $f(\varepsilon) = (e^{\varepsilon/k_{\rm B}T}+1)^{-1}$ is the Fermi–Dirac distribution and μ is the chemical potential. At $T = 0$ in a gapped insulating phase, this reduces to $\sigma_{xy} = (e^2/h)\,C$ where $C = C_\uparrow + C_\downarrow$ is the total Chern number [39].

B. Anomalous Nernst conductivity

The intrinsic ANC, relating the transverse electric current to a longitudinal temperature gradient via $J_y = \alpha_{xy}(-\partial_x T)$, is obtained from the Berry curvature weighted by the entropy kernel. Using integration by parts and the identity $\int_\varepsilon^\infty f(\varepsilon'-\mu)d\varepsilon' = k_{\rm B}T\ln[1+e^{(\mu-\varepsilon)/k_{\rm B}T}]$, we arrive at the exact result [32, 40]:

$$\alpha_{xy} = \frac{ek_{\rm B}}{\hbar}\sum_\sigma\int\frac{d^2k}{(2\pi)^2}\,\Omega_-^{z,\sigma}(\boldsymbol{k})\,\ln\left[1+e^{(\mu-\varepsilon_-^\sigma(\boldsymbol{k}))/k_{\rm B}T}\right]. \tag{13}$$

Equation (13) has the important property that $\alpha_{xy} \to 0$ as $T \to 0$ in any gapped insulating phase, because the logarithm becomes either $(\mu-\varepsilon_-^\sigma)/k_{\rm B}T \to +\infty$ (filled bands, exponentially cut off by the gap) or approaches zero (empty bands). Consequently, α_{xy} is entirely thermally activated.

C. Anomalous thermal Hall conductivity

The ATHC, defined by $J_y^Q = -\kappa_{xy}^{(0)}\,\partial_x T$, is given by the c_2 function formula derived via Luttinger's gravitational potential approach [41, 42]:

$$\kappa_{xy}^{(0)} = -\frac{k_{\rm B}^2 T}{\hbar}\sum_\sigma \int \frac{d^2k}{(2\pi)^2}\,\Omega_-^{z,\sigma}(\boldsymbol{k})\,c_2\big(f(\varepsilon_-^\sigma(\boldsymbol{k})-\mu)\big)\,, \tag{14}$$

where the function c_2 is defined as

$$c_2(f) = \int_0^f \left(\ln\frac{1-t}{t}\right)^2 dt, \quad c_2(0)=0, \quad c_2(1)=\frac{\pi^2}{3}. \tag{15}$$

The value $c_2(1) = \pi^2/3$ can be established precisely using the Dirichlet eta function: $c_2(1) = 2\int_0^\infty u^2 e^u/(e^u+1)^2\,du = 4\eta(2) = \pi^2/3$. In the low-temperature limit with the chemical potential in the gap, $f \to 1$ for all occupied states and $c_2(f) \to \pi^2/3$, gives

$$\left.\frac{\kappa_{xy}^{(0)}}{T}\right|_{T\to 0} = -\frac{\pi^2 k_{\rm B}^2}{3\hbar}\cdot\frac{C}{2\pi} = -\frac{\pi^2 k_{\rm B}^2}{3h}\,C. \tag{16}$$

D. Wiedemann–Franz law for anomalous transport

Comparing Eq. (16) with $\sigma_{xy} = e^2 C/h$ at $T = 0$, we obtain the anomalous Wiedemann-Franz law: [43]

$$\frac{\kappa_{xy}^{(0)}}{\sigma_{xy}} = -\frac{\pi^2 k_{\rm B}^2}{3e^2}\,T \equiv -L_0 T, \tag{17}$$

where $L_0 = \pi^2 k_{\rm B}^2/3e^2 \approx 2.44\times 10^{-8}\,{\rm W\,\Omega\,K^{-2}}$ is the Lorenz number. This relation holds exactly in any topological phase at $T \to 0$ and serves as an unambiguous experimental test of the Berry phase origin of the thermal Hall response.

E. Mott formula for the Nernst conductivity

At low temperature $k_{\rm B}T \ll \mu$, the Sommerfeld expansion gives [44]

$$\alpha_{xy} \approx -\frac{\pi^2 k_{\rm B}^2 T}{3e}\left.\frac{\partial\sigma_{xy}(\varepsilon)}{\partial\varepsilon}\right|_{\varepsilon=\mu}, \tag{18}$$

the anomalous analogue of the Mott formula. Equation (18) shows that α_{xy} is proportional to the energy derivative of the AHC, making it sensitive to the sharpness of Berry curvature features near the Fermi level.

F. Spin-resolved transport coefficients

Since the AM effective Hamiltonian is block diagonal in spin, we define spin resolved versions of all three coefficients. The spin Hall conductivity

$$\sigma_{xy}^s = \frac{e}{2}\int\frac{d^2k}{(2\pi)^2}\left[\Omega_-^{z,\uparrow} f_-^\uparrow - \Omega_-^{z,\downarrow} f_-^\downarrow\right], \tag{19}$$

the spin Nernst conductivity

$$\alpha_{xy}^s = \frac{\hbar k_{\rm B}}{2}\int\frac{d^2k}{(2\pi)^2}\Big[\Omega_-^{z,\uparrow}\ln\Big(1+e^{(\mu-\varepsilon_-^\uparrow)/k_{\rm B}T}\Big) - \Omega_-^{z,\downarrow}\ln\Big(1+e^{(\mu-\varepsilon_-^\downarrow)/k_{\rm B}T}\Big)\Big], \tag{20}$$

and the spin thermal Hall conductivity

$$\kappa_{xy}^{(0),s} = -\frac{\hbar k_{\rm B}^2 T}{2e}\int\frac{d^2k}{(2\pi)^2}\left[\Omega_-^{z,\uparrow} c_2(f_-^\uparrow) - \Omega_-^{z,\downarrow} c_2(f_-^\downarrow)\right]. \tag{21}$$

In the QSH phase where $\Omega_-^{z,\uparrow} = -\Omega_-^{z,\downarrow}$ and $\varepsilon_-^\uparrow = \varepsilon_-^\downarrow$, the total $\sigma_{xy} = \alpha_{xy} = \kappa_{xy}^{(0)} = 0$ but both σ_{xy}^s and $\kappa_{xy}^{(0),s}$ are nonzero, providing a route to pure spin caloritronic effects.

G. Symmetry of transport coefficients under θ

The polarization angle dependence of all three anomalous coefficients follows from the symmetry $j_1(-\theta) = j_1(\theta)$ and $j_2(-\theta) = j_2(\theta)$ (even in θ) combined with the exchange symmetry $j_1 \leftrightarrow j_2$ under $\theta \to \pi/2-\theta$. We can show that

$$X(\pi/2-\theta) = -X(\theta), \quad X \in \{\sigma_{xy}, \alpha_{xy}, \kappa_{xy}^{(0)}\}, \tag{22}$$

which requires $X(\pi/4) = 0$, i.e., all anomalous transport coefficients vanish at the diagonal polarization angle where $\mathcal{M}_{xy}$ symmetry is preserved. Additionally, $|X|$ is maximized when $\theta = 0$ or $\theta = \pi/2$ (LPL along a crystal axis), and all three coefficients share the same d-wave nodal pattern in the (A_0, θ) plane.

V. RESULTS AND DISCUSSION

We consider a pristine system and thus our results describe intrinsic contributions; extrinsic effects may modify magnitudes but not symmetry or signatures of studied phases. In this study, we explain the Floquet-engineered altermagnetic system using a non-interacting band framework. There are various arguments to support this approximation. First, the anomalous Hall, Nernst, and thermal Hall responses taken into consideration here are intrinsic transport coefficients that are mostly governed by the underlying band structure's Berry curvature, which is well-represented in an effective single-particle picture. Second, as long as the driving frequency is higher than the characteristic interaction and bandwidth scales, electron-electron interactions in the high-frequency (off-resonant) Floquet regime primarily result in weak renormalizations of the band parameters and

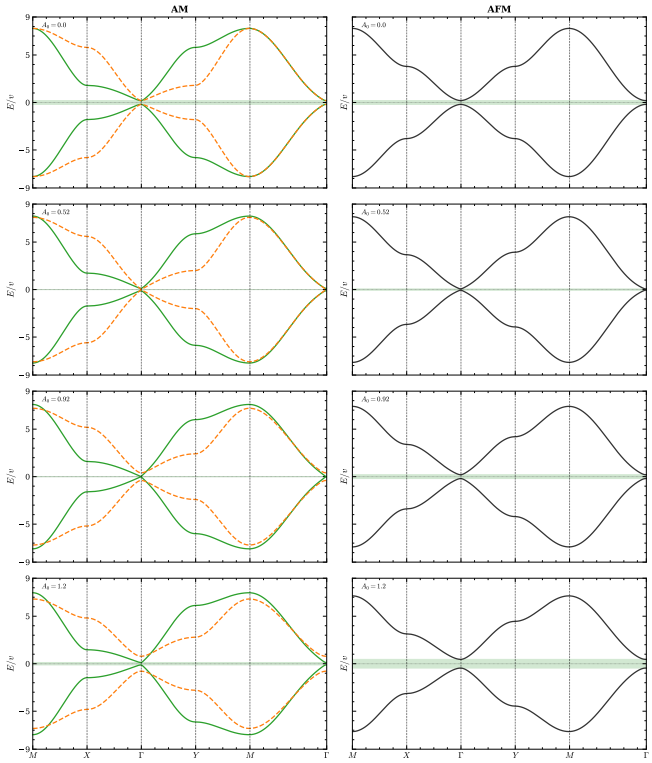

FIG. 1. Energy dispersions and topological phase diagram of the QSH AM under LPL. Spin-resolved band structure of the QSH AM ($m = 3.8v$, $b = -v$, $t_a = \sqrt{3}$) at $A_0 = 0.0, 0.3, 0.7, 1.0$ ($\theta = 0$). Spin-up (solid teal) and spin-down (dashed orange) bands evolve differently; the spin-down gap closes at A_0^{c1} producing the Chern phase, while the spin-up gap closes at A_0^{c2}.

do not qualitatively change the topology of the effective Hamiltonian. Third, the main phenomena discussed here, the symmetry-controlled angular dependence, spin, selective gap closings, and Chern number changes, are resistant against weak interaction effects because they are determined by symmetry and band topology. Lastly, investigations and first-principles research show that a quasiparticle description is still viable for candidate altermagnetic materials like RuO_2 and MnTe-based systems, supporting the use of an effective non-interacting model.

A. Band Structures

The spin-resolved band structures of the QSH altermagnetic (AM) and QSH antiferromagnetic (AFM) insulators are examined for four values of the LPL amplitude and results are shown in Fig. 1. In the QSH AM case, the spin-up and spin-down bands evolve differently with increasing A_0: the spin-down gap closes at $A_0^{c1} \approx 0.521$ while the spin-up gap remains open, leading to the emergence of an intermediate Chern phase. Upon further increasing the amplitude to $A_0^{c2} \approx 0.918$, the spin-up gap also closes, and the system transitions

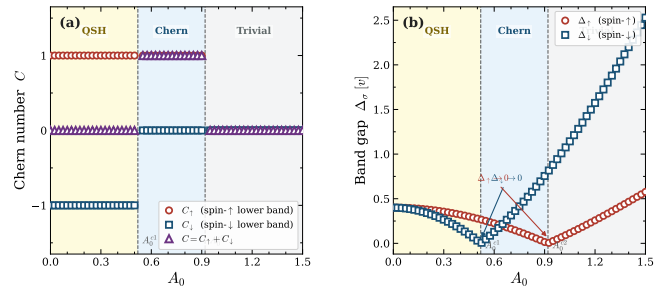

FIG. 2. Topological phase transition (a) Spin Chern numbers $C_\uparrow$, $C_\downarrow$, total C as a function of the A_0. The Chern number C in each panel is calculated by integrating the total Berry curvature. (b) Band gap as a function of the A_0. At two different values of amplitude the band gap vanishes separately for spin-up and spin-down that shows two topological phase transitions. For the trivial AM the Chern number remains $C = 0$ throughout. For the QSH AM, C evolves from 0 (QSH phase, $A_0 = 0.0, 0.3$) to +1 (Chern phase, $A_0 = 0.6$) to +2 ($A_0 = 0.9$), reflecting the sequential topological transitions driven by the differential spin-sector renormalization $j_1 \neq j_2$.

into a trivial phase. In contrast, for the QSH AFM case, the spin-up and spin-down bands remain perfectly degenerate for all values of A_0, and both gaps close simultaneously at a single critical amplitude, resulting in the absence of any intermediate Chern phase. This contrast is made explicit in phase diagram which shows the topological phase transitions from QSH topological insulator to Chern insulator and then to Trivial altermagnetic insulator. The spin-resolved Chern numbers evolve continuously with the amplitude A_0, revealing two critical values: $A_0^{c1} \approx 0.521$ marking the transition from the quantum spin Hall (QSH) phase to the Chern insulating phase, where the spin-down gap closes, and $A_0^{c2} \approx 0.918$ marking the transition from the Chern phase to the trivial phase, where the spin-up gap closes. The total Chern number $C = C_\uparrow + C_\downarrow$ and the spin Chern number $C_s = (C_\uparrow - C_\downarrow)/2$ thus identify three distinct phases as shown in Fig. 2(a). This sequential gap-closing mechanism arises because the linearly polarized light (LPL) renormalizes the two spin blocks differently due to the condition $j_1 \neq j_2$, causing the spin-down gap to vanish first, Fig 2(b). Such a sequence is entirely absent in the antiferromagnetic (AFM) case, where the preserved $\mathcal{PT}$ symmetry enforces $\Delta_\uparrow = \Delta_\downarrow$ for all A_0.

B. Berry curvature maps and topological phase transitions

For the trivial AM, the Berry curvature distribution shifts and concentrates as A_0 increases, but the Chern number remains $C = 0$ throughout since no gap closing occurs. The gradual transfer of Berry curvature weight between the Γ and X/Y regions reflects the LPL induced renormalization of the spin dependent hopping amplitudes, $j_1 = J_0(A_0) \to 0$ while $j_2 = J_0(0) = 1$ is un-

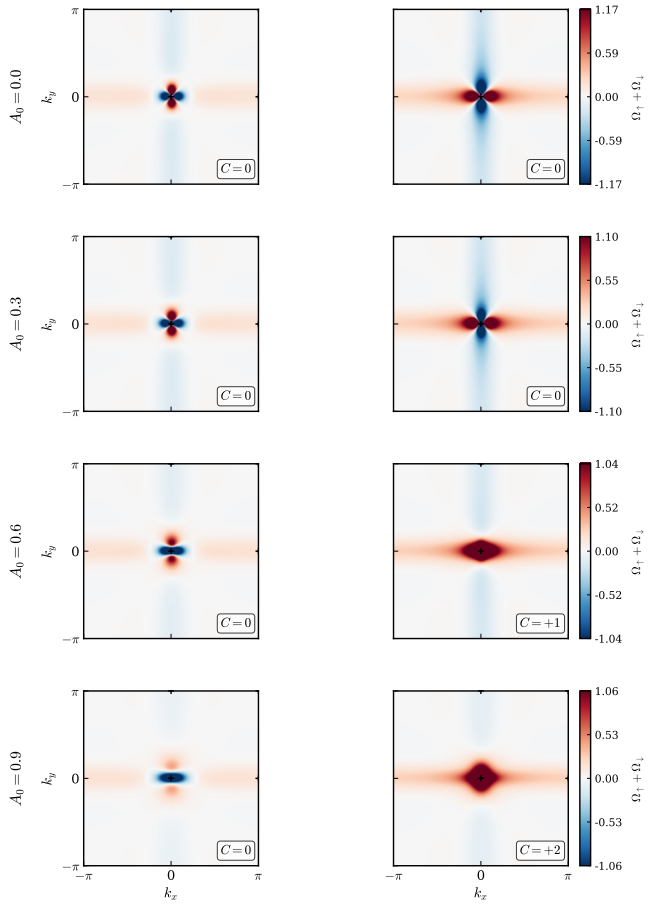

FIG. 3. Total occupied Berry curvature $\Omega_-^\uparrow(\boldsymbol{k})+\Omega_-^\downarrow(\boldsymbol{k})$ across the Brillouin zone. (Left panels) Trivial AM ($m = 4.2v$, $b = -v$, $t_a = \sqrt{3}$). (Right panels) QSH AM ($m = 3.8v$, $b = -v$, $t_a = \sqrt{3}$). Rows from top to bottom: $A_0 = 0.0$, 0.3, 0.6, 0.9 at $\theta = 0$. The Chern number C in each panel is calculated by integrating the total Berry curvature. For the trivial AM the Chern number remains $C = 0$ throughout. For the QSH AM, C evolves from 0 (QSH phase, $A_0 = 0.0$, 0.3) to +1 (Chern phase, $A_0 = 0.6$) to +2 ($A_0 = 0.9$), reflecting the sequential topological transitions driven by the differential spin-sector renormalization $j_1 \neq j_2$.

changed for $\theta = 0$.

For the QSH AM, the initial $A_0 = 0$ configuration has $C = 0$ (QSH phase with $C_\uparrow = -1$, $C_\downarrow = +1$). At $A_0 = 0.6$ the Berry curvature develops a noticeable asymmetry between spin sectors, the Chern number jumps to $C = +1$, making the entry into the Chern insulating phase. Further increasing to $A_0 = 0.9$ gives $C = +2$, consistent with the sequential transitions we identify numerically. The occupied Berry curvature, shown in Fig. 3, underlies these behaviors and reflects the sequential topological transitions driven by the differential renormalization of the spin sectors ($j_1 \neq j_2$).

C. Anomalous Hall conductivity

At zero temperature, the anomalous Hall conductivity shows distinct behavior depending on the altermagnetic phase as shown in Fig. 4(a-c). In the trivial altermagnetic phase (panel 4(a)), no topological transition occurs, yet increasing the staggered potential strength A_0 leads to a nontrivial energy dependence as the broken $C_{4z}\mathcal{T}$ symmetry allows Berry curvature to accumulate. In contrast, the quantum spin Hall altermagnetic phase in panel 4(b) shows that as A_0 increases through the first critical value A_0^{c1}, a quantized plateau $\sigma_{xy} = +e^2/h$ emerges within the gap region, signaling a Chern insulating phase. Notice that the Berry curvature sharpens near gap closing and thus the physical quantities show peak at this location. The conductivity satisfies the exact even symmetry $\sigma_{xy}(E_F) = \sigma_{xy}(-E_F)$, as shwon in panel 4(c) confirmed by a residual root-mean-square error of $7.1 \times 10^{-17}\, e^2/h$, which reflects a particle–hole-like symmetry of the Berry curvature. Finally, tracking the spin-resolved Chern numbers and total σ_{xy} as functions of A_0 unambiguously identifies three phases: quantum spin Hall ($C = 0$), Chern ($C = -1$), and trivial ($C = 0$), with critical amplitudes $A_0^{c1} = 0.521$ and $A_0^{c2} = 0.918$ as shown in panel 4(d).

D. Anomalous Nernst conductivity: thermal activation and Mott scaling

Figure 5 shows the temperature evolution of the anomalous Nernst coefficient (ANC). For the trivial altermagnet, panel 5(a), it shows that the ANC vanishes as temperature approaches zero, confirming the thermally activated character predicted by theory, as the effect cancels in filled bands at zero temperature due to cancellation of Berry curvature contributions weighted by the entropy kernel. At low temperatures, plotting $\alpha_{xy}/(k_\mathrm{B}T)$ in panel 5(b) versus Fermi energy reveals convergence to the Mott prediction $-\frac{\pi^2 k_\mathrm{B}}{3e}\frac{\partial\sigma_{xy}}{\partial\varepsilon}\big|_{\varepsilon=\mu}$, with the root-mean-square deviation decreasing as temperature lowers. Increasing the light amplitude A_0 both enhances the magnitude of the ANC and shifts its peak position, because linearly polarized light modifies the Berry curvature distribution and hence the energy derivative of the anomalous Hall con as shwon in panel 5(c). The ANC satisfies the exact odd symmetry $\alpha_{xy}(E_F) = -\alpha_{xy}(-E_F)$ in panel 5(d). The temperature dependence of the ANC at fixed Fermi energies for both trivial and quantum spin Hall altermagnets shows a characteristic non-monotonic behavior: it rises from zero at $T = 0$, peaks at a characteristic temperature T^*, and then decreases at high temperature as thermal broadening averages over Berry curvature features of opposite signs. The peak temperature T^* is tunable by light intensity; for the trivial altermagnet at a given Fermi energy, $k_\mathrm{B}T^*$ increases with A_0, while for the quantum spin Hall altermagnet T^* is even more strongly A_0-dependent. This tunability provides

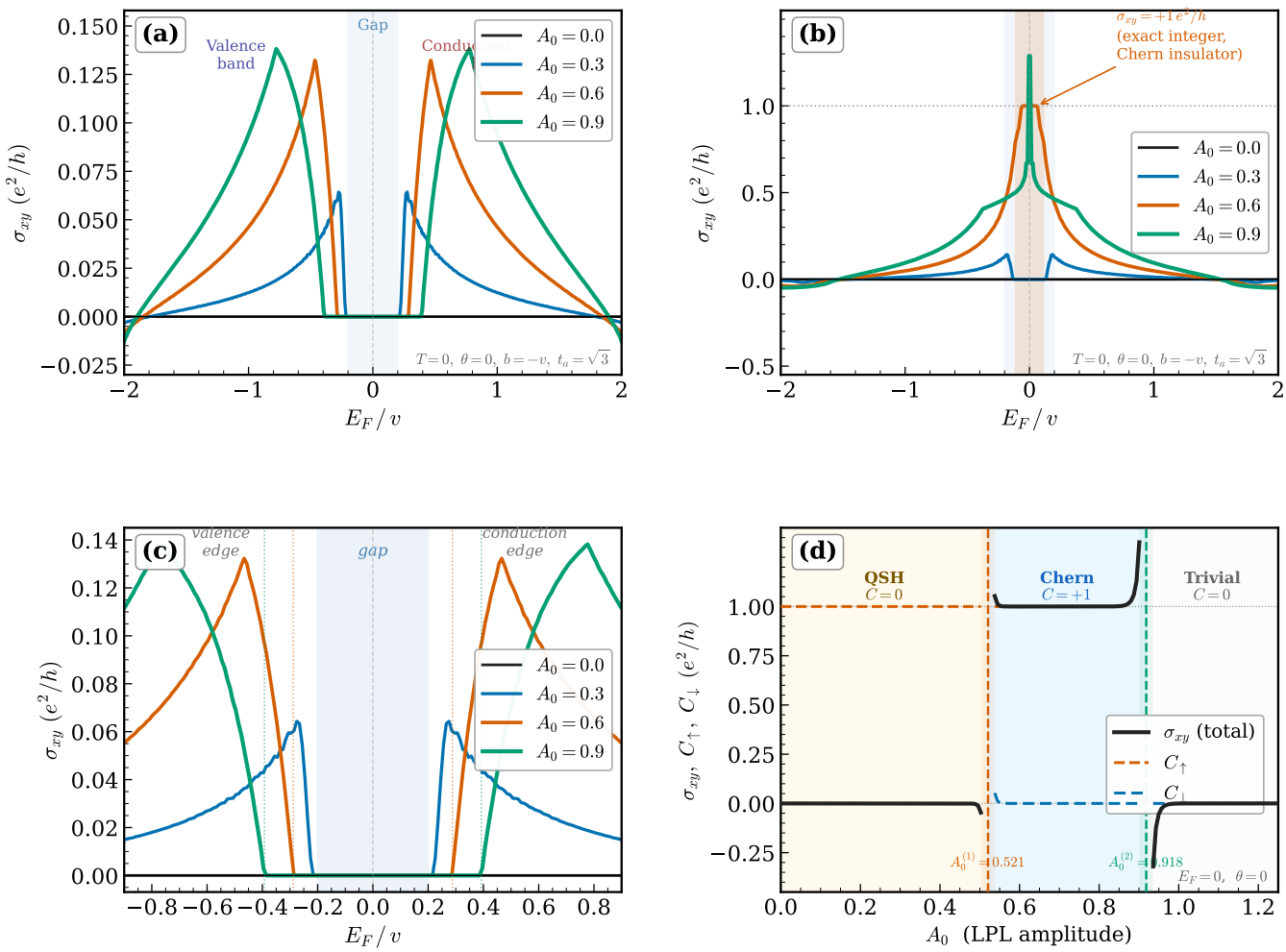

FIG. 4. (a-c) Anomalous Hall conductivity σ_{xy} at $T = 0$ as a function of Fermi energy. plot in (a) shows σ_{xy} for the trivial AM; even though no topological transition occurs, σ_{xy} develops a non-trivial energy dependence with increasing A_0 as the broken $C_{4z}\mathcal{T}$ symmetry allows Berry curvature to accumulate. Results in (b) shows the QSH AM; as A_0 increases through A_0^{c1}, a quantized plateau $\sigma_{xy} = +e^2/h$ emerges in the gap region, characteristic of the Chern insulating phase. The approach to the plateau is confirmed by the transition value shown by the dashed line. panel (c) verifies the exact even symmetry $\sigma_{xy}(E_F) = \sigma_{xy}(-E_F)$ with residual RMS error $7.1 \times 10^{-17}\, e^2/h$, confirming a symmetry $\varepsilon \to -\varepsilon$ (particle–hole-like) of the Berry curvature in this model. (d) Tracks the spin-resolved Chern numbers and the total σ_{xy} as a function of A_0, unambiguously identifying the QSH ($C = 0$), Chern ($C = -1$), and trivial ($C = 0$) phases with critical amplitudes $A_0^{c1} = 0.521$ and $A_0^{c2} = 0.918$.

a direct experimental knob for thermoelectric optimization.

E. Temperature and chemical potential dependence

Figure 6(a-c) show all three physical conductivities which share an identical nodal structure with zeros at $\theta = \pm\pi/4$ (where $\mathcal{M}_{xy}$ symmetry is restored) and extrema near $\theta = 0, \pm\pi/2$. The sign reversal under $\theta \to -\theta$ is visible, and the magnitude scales with A_0. Furthermore, the sign of all three coefficients can be controlled by rotating the polarization direction, providing a purely optical route to reversing the thermal Hall response a significant advantage over approaches requiring magnetic field reversal. Importantly, this entire polarization pattern is absent in the conventional AFM, where $\sigma_{xy} = \alpha_{xy} = \kappa_{xy}^{(0)} = 0$ for all θ, making the angle-dependent thermal transport a definitive experimental fingerprint of altermagnetism.

F. Intrinsic Nernst and thermal Hall in the AM insulator

Figure 7 shows the intrinsic Nernst $\alpha_{xy}^{\rm in}$ and thermal Hall $\kappa_{xy}^{\rm in}$ coefficients as functions of chemical potential for the trivial AM insulator at three LPL amplitudes. The Nernst coefficient panels 7(a-c) exhibits a characteristic double-peak structure with opposite signs across the gap,

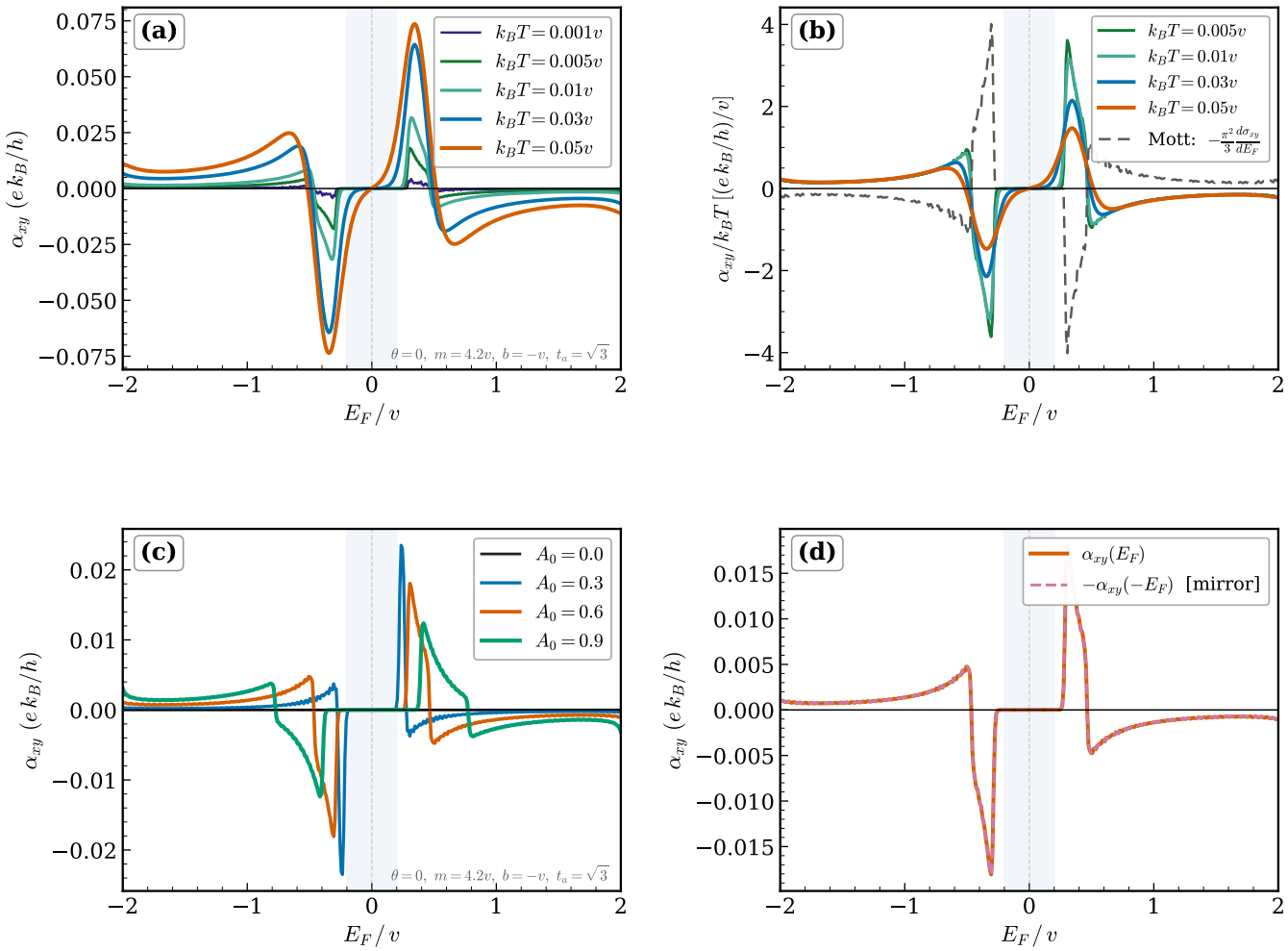

FIG. 5. Anomalous Nernst conductivity α_{xy} as a function of Fermi energy. In all panels the trivial AM is considered by given $m = 4.2v$, $b = -v$, $t_a = \sqrt{3}$, $\theta = 0$. (a) at $A_0 = 0.6$ for six temperatures, demonstrating the thermally activated nature ($\alpha_{xy} \to 0$ as $T \to 0$). (b) Scaled quantity $\alpha_{xy}/(k_\mathrm{B}T)$ as a function of E_F at $A_0 = 0.6$; the Mott prediction $-(\pi^2 k_\mathrm{B}/3e)\,\partial\sigma_{xy}/\partial\varepsilon|_\mu$ (dashed) is approached as $T \to 0$. (c) at $k_BT = 0.005v$ for four LPL amplitudes, showing the evolution of the Nernst response with light intensity. (d) Odd symmetry $\alpha_{xy}(E_F) = -\alpha_{xy}(-E_F)$ verified for $A_0 = 0.6$; Notice that a RMS error is $5.8 \times 10^{-17}\, ek_\mathrm{B}/h$.

whose magnitude grows with A_0. The thermal Hall coefficient panels 7(d-f) shows a single negative peak inside the gap that deepens as A_0 increases. At $A_0 = 0.9$ the peak value reaches approximately $-0.14\, e^2 k_\mathrm{B}T/h$, about 2.3 times larger than at $A_0 = 0.3$. The insulating gap is highlighted in green; within it, $\alpha_{xy}^{\mathrm{in}} \to 0$ as $T \to 0$ (consistent with our analytical prediction) while $\kappa_{xy}^{\mathrm{in}}$ remains finite, tracking the WF prediction.

G. Transport Signatures Across All Topological Phases

We commence by examining the band structure induced by LPL as a foundation for understanding the transport signatures. Figure 8 shows the spin-resolved band structures and the corresponding constant energy contours of the LPL driven d-wave altermagnetic quantum spin Hall system. Solid curves represent the spin-up sector, while dashed curves correspond to the spin-down sector. The parameters used are altermagnetic parameters and these are fixed in all phases, $m = 3.8v, b = -v, t_a = \sqrt{3}$ for these specified values of AM parameters the system is in equilibrium and satisfies $|m| < |b|(1+t_a^2)$. Since $|b|(1 + t_a^2) = 4v$, the undriven or weakly driven system sits comfortably within the quantum spin Hall regime.

Figure 8(a) presents the band structure in the quantum spin Hall phase. The bulk gap remains open throughout this regime, as the inset makes clear, yet the valence and conduction bands are inverted relative to the trivial case. The spin-up and spin-down sectors carry opposite Chern numbers, $C_\uparrow = -1$ and $C_\downarrow = +1$ so the total Chern number vanishe, while the spin Chern number remains finite, $C_s \sim C_\uparrow - C_\downarrow \neq 0$. This confirms that the system is a QSH insulator rather than a trivial one. The open

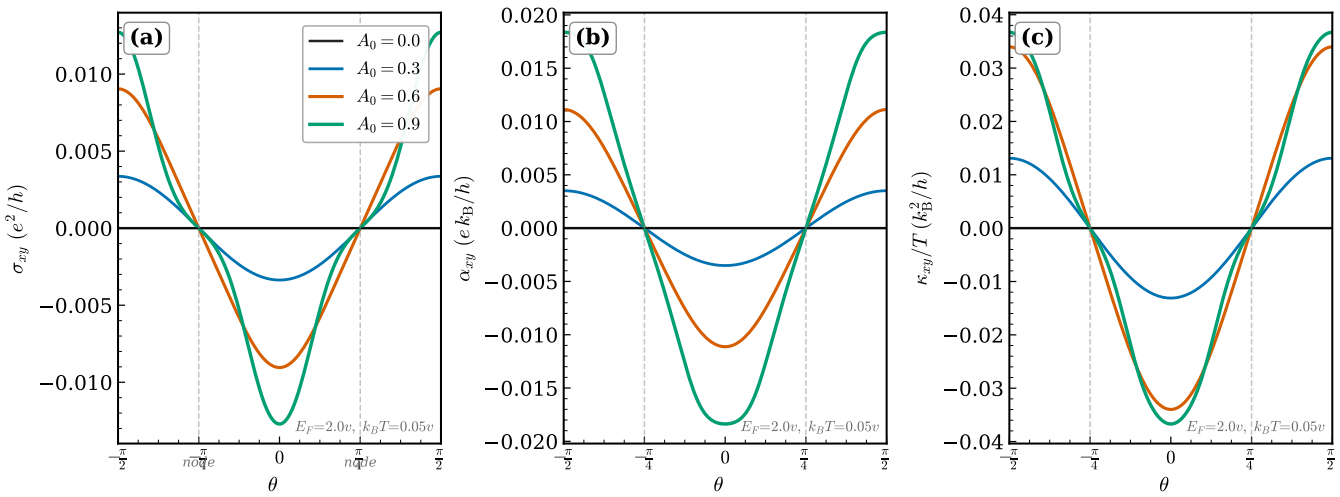

FIG. 6. Physical conductivities as a function of the Polarization angle θ. All three conductivities (a-c) show a d-wave nature under LPL that is a remarkable result and a fingerprint for AM.

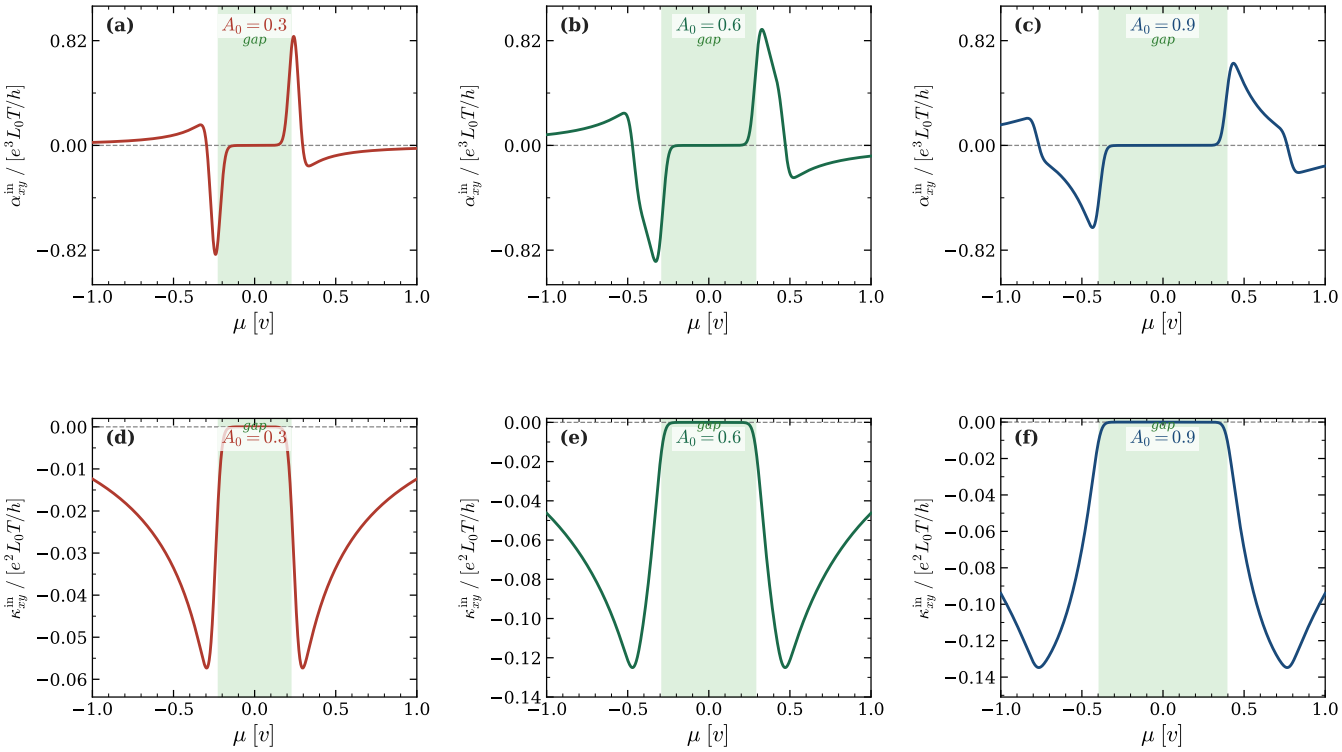

FIG. 7. Nernst and thermal conductivities in terms of chemical potential (a-c) Nernst conductivity as a function of chemical potential at different values of amplitude. (d-f) Thermal Hall conductivity as a function of chemical potential at different values of amplitude.

gap (visible in the inset) signals insulating behaviour, and the band inversion signals the underlying nontrivial topology.

Figure 8(b) shows the first light-induced Chern insulating phase, which carries a total Chern number $C = -1$. With increase in light amplitude A_0, the Bessel-renormalized hopping terms redirect the effective masses of the two spin sectors in different directions. Consequently, one spin gap closes and reopens before the other. Here the spin-down sector turns topologically trivial first, while the spin-up sector remains inverted, $C_\uparrow = -1$ and $C_\downarrow = 0$ that results in total chern number is -1.

The inset captures the local gap closing associated with this topological phase transition that is a necessary fea-

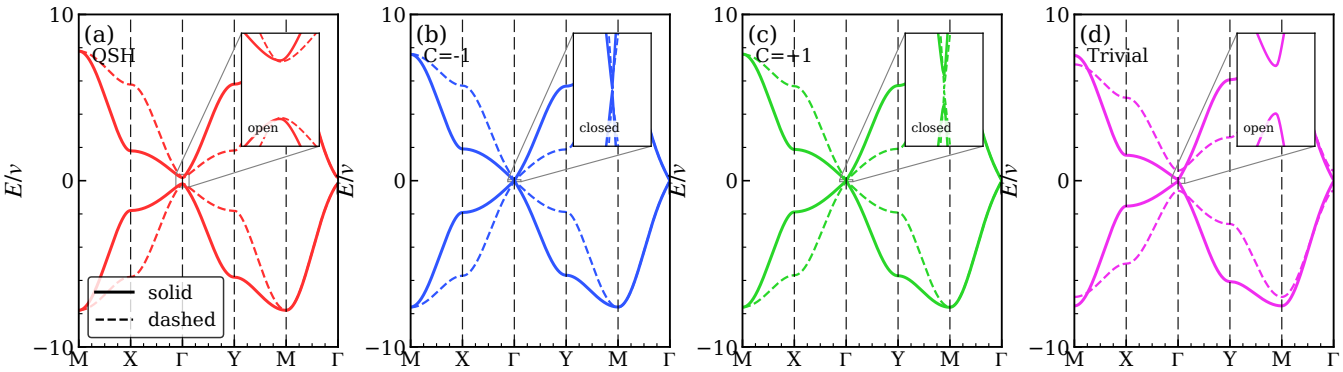

FIG. 8. Band structures and topological phase diagram of the QSH AM under LPL. (a)–(d): Spin-resolved band structure of the QSH AM ($m = 3.8v$, $b = -v$, $t_a = \sqrt{3}$) at $A_0 = 0.0, 0.3, 0.7, 1.0$ ($\theta = 0$). Spin-up (solid teal) and spin-down (dashed orange) bands evolve differently; the spin-down gap closes at A_0^{c1} producing the Chern phase, while the spin-up gap closes at A_0^{c2}. Insets show $\Delta_\uparrow$ and $\Delta_\downarrow$. Lower panels: spin Chern numbers $C_\uparrow$, $C_\downarrow$, total C, and spin Chern number $C_s = (C_\uparrow - C_\downarrow)/2$ vs. A_0.

ture, since a change in Chern number can only occur when the bulk gap closes and then reopens. The result is a spin-polarized Chern insulator whose nonzero Chern number originates predominantly from a single spin channel.

Figure 8(c) illustrates the complementary spin-polarized Chern phase with total Chern number $C = +1$. Now it is the other spin sector that remains topologically active, while the first has already become trivial, $C_\uparrow = 0$ and $C_\downarrow = +1$ so that the total Chern number is $+1$.

The coexistence of both $C = -1$ and $C = +1$ phases makes it clear that the sign of the light-induced Chern number can be controlled through the polarization direction combined with the anisotropic Floquet renormalization. This tunability is a distinguishing feature of the altermagnetic system, where linearly polarized light breaks the symmetry that would otherwise relate the two spin sectors.

Figure 8(d) shows the trivial insulating phase reached at larger light amplitude. Once the remaining inverted spin sector also undergoes its own gap closing and reopening, both sectors become topologically trivial, $C_\uparrow = 0, C_\downarrow = 0$

Although the inset confirms that the bulk gap is still open, the phase has lost its topological character entirely because the band inversion is gone. This panel therefore represents an ordinary trivial insulator, and the full sequence of phase transitions reads

$$\text{QSH} \to C = -1 \text{ or } C = +1 \to \text{Trivial}. \tag{23}$$

A particularly illuminating way to present our results is to compare the three anomalous transport coefficients; the AHC σ_{xy}, ANC α_{xy}, and ATHC $\kappa_{xy}^{(0)}/T$ across all four distinct topological phases that the altermagnetic system can have under LPL irradiation: the QSH phase ($C = 0$), the two Chern insulating phases ($C = -1$ and $C = +1$), and the trivial phase ($C = 0$). This phase comparison, reveals how each transport coefficient encodes a distinct topological fingerprint and how the Berry curvature redistribution in the Brillouin zone manifests differently in charge, thermoelectric, and thermal channels.

Figure 9(a-d) presents σ_{xy} as a function of Fermi energy E_F for representative parameter choices in each of the four topological phases. The top panel overlays all four phases on a common axis, making the contrast immediately apparent. The QSH phase, panel 9(a), displays $\sigma_{xy} \approx 0$ throughout the gap region, as expected for a $\mathcal{PT}$-broken but $C_{4z}\mathcal{T}$-symmetric state where $C_\uparrow = -C_\downarrow$ and the Berry curvatures of the two spin sectors cancel. Outside the gap, σ_{xy} acquires small but nonzero values as the Fermi level probes bands with unequal spin resolved Berry curvature densities.

The two Chern insulating phases shows different behaviour. For the $C = -1$ phase, panel 9(b) ($A_0 = 0.625$, $\theta = 0.829\,\text{rad}$, $\Delta = 0.002v$), σ_{xy} develops a sharp negative dip near $E_F = 0$, approaching $-e^2/h$ as the Fermi level sits in the tiny gap. For the $C = +1$ phase, panel 9(c), ($A_0 = 0.625$, $\theta = 0.742\,\text{rad}$, $\Delta = 0.002v$), the sign is reversed and σ_{xy} rises sharply toward $+e^2/h$ within the gap. The near quantized values in panels 9(b) and 9(c) reflect the fact that $\sigma_{xy} = Ce^2/h$ in the low temperature insulating limit, with the sign controlled entirely by the polarization angle θ. The ability to switch between $C = -1$ and $C = +1$ by rotating the polarization direction, without any change in material or magnetic field is one of the most striking features of Floquet engineered altermagnetism.

The trivial phase, panel 9(d), ($A_0 = 1.075$, $\theta = 0$, $\Delta = 0.137v$) shows a broad, positive σ_{xy} that peaks near the band edges and decays away from the gap. This finite but non quantized response arises from the residual Berry curvature of the trivially gapped bands, and has no topological protection. The contrast between the sharp, sign changing σ_{xy} in the Chern phases and the broad, featureless response in the trivial phase provides a clear experimental discriminator between topological and non-

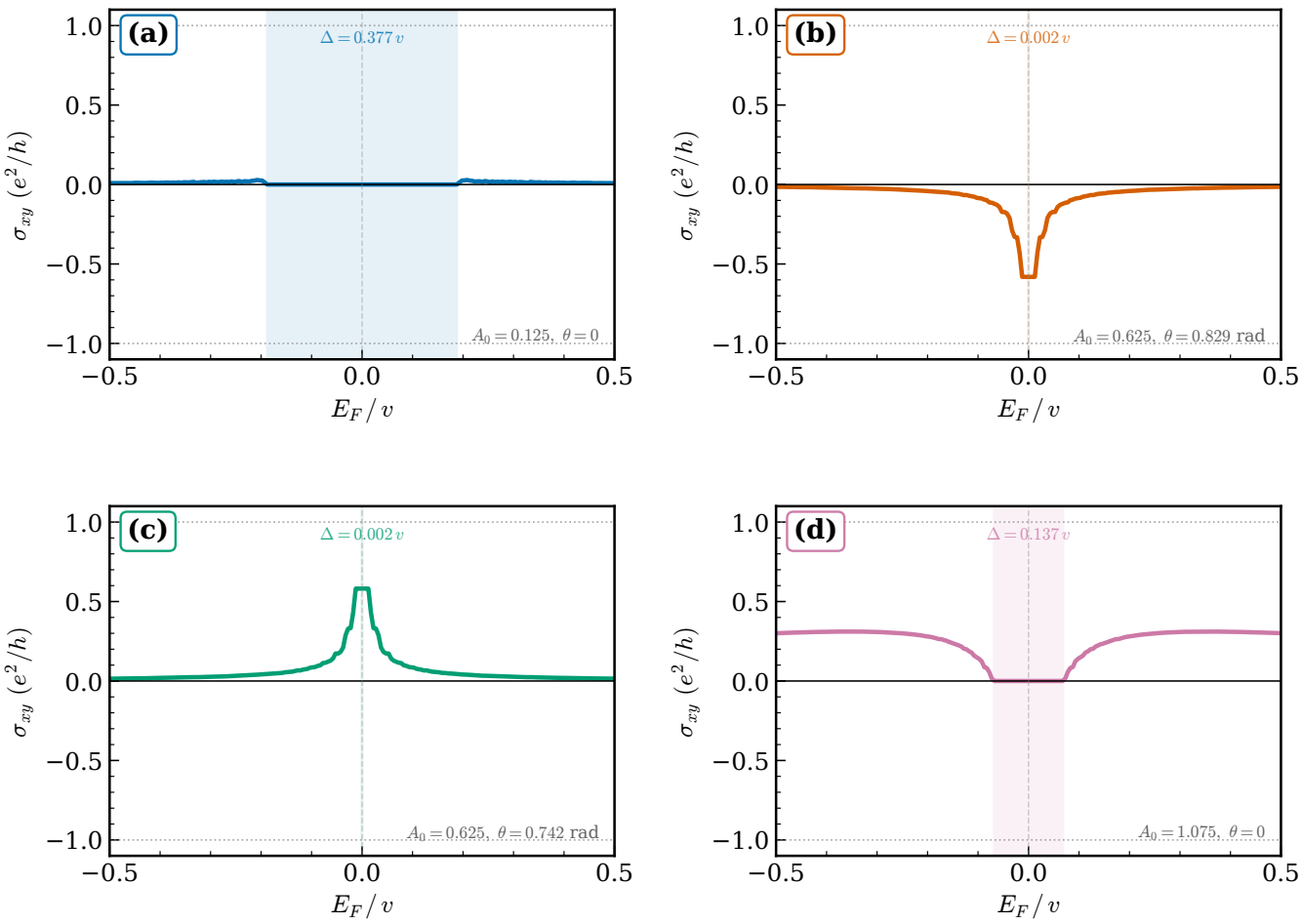

FIG. 9. Anomalous Hall conductivity σ_{xy} across all four topological phases. Top panel: overlay of all four phases as a function of Fermi energy E_F/v. Dotted horizontal lines mark $\pm e^2/h$. Individual panels: (a) QSH phase ($C = 0$, $A_0 = 0.125$, $\theta = 0$, $\Delta = 0.377v$); (b) Chern phase ($C = -1$, $A_0 = 0.625$, $\theta = 0.829\,\mathrm{rad}$, $\Delta = 0.002v$); (c) Chern phase ($C = +1$, $A_0 = 0.625$, $\theta = 0.742\,\mathrm{rad}$, $\Delta = 0.002v$); (d) trivial phase ($C = 0$, $A_0 = 1.075$, $\theta = 0$, $\Delta = 0.137v$). Blue shading marks the insulating gap in each panel. The Chern phases develop near-quantized values $\sigma_{xy} \to Ce^2/h$ in the gap, while the QSH and trivial phases show vanishing and non-quantized responses, respectively.

topological regimes.

Figure 10(a-d) shows α_{xy} at $k_BT = 0.01v$ for the same four phases. Since $\alpha_{xy}(T = 0) \equiv 0$ in any gapped phase, all curves are shown at a small but finite temperature to reveal the thermally activated structure.

The QSH phase, panel 10(a), displays a very small α_{xy} with a peak value of only $+0.015\,ek_B/h$, concentrated near the gap edges. This small magnitude reflects the nearly complete cancellation of Berry curvature contributions from the two spin sectors in the QSH state: the Nernst response is thermally activated by band edge excitations across a relatively large gap $\Delta = 0.377v$, and the partial cancellation suppresses the signal further.

The Chern phases exhibit dramatically enhanced Nernst responses. For $C = -1$, panel 10(b), α_{xy} develops a pronounced bipolar structure near $E_F = 0$, with a positive peak of $+1.882\,ek_B/h$ and a negative trough of comparable magnitude, separated by the tiny gap $\Delta = 0.002v$. The $C = +1$ phase, panel 10(c) shows the mirror image of this structure, with a peak of $+1.883\,ek_B/h$ of opposite internal asymmetry. These large values, more than 100 times the QSH response arise because the extremely small gaps in the Chern phases ($\Delta \approx 0.002v$) allow thermal excitation across the gap even at $k_BT = 0.01v$, and the Berry curvature is sharply concentrated near the gap closing $\boldsymbol{k}$ points. The Nernst response is therefore a far more sensitive probe of topological gap size than the Hall conductivity: while σ_{xy} approaches $\pm e^2/h$ in both Chern phases, α_{xy} distinguishes them by its magnitude and internal lineshape.

The trivial phase, panel 10(d) gives a small, broad Nernst response with peak $+0.102\,ek_B/h$, confined to the band edges and showing no sign change within the gap. This is qualitatively different from both Chern phases and provides a clear thermal fingerprint of the trivial insulating state.

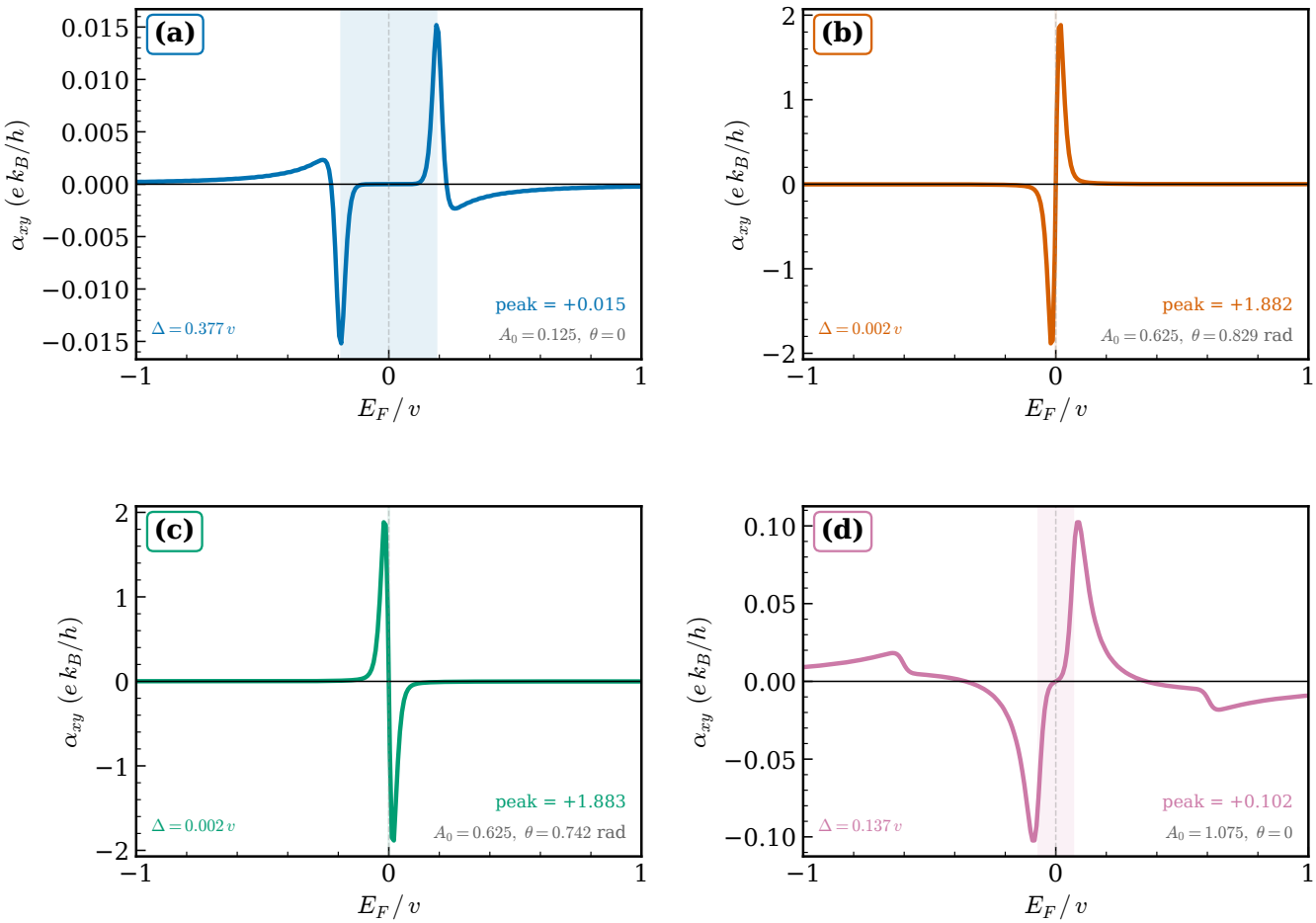

FIG. 10. Anomalous Nernst conductivity α_{xy} across all four topological phases at $k_BT = 0.01v$. Top panel: overlay of all four phases. Individual panels with the same parameter choices as Fig. 9. Noice that the dramatic enhancement in the Chern phases (b-c): peak values of $\approx 1.88\, ek_B/h$ compared to $0.015\, ek_B/h$ in the QSH phase (a), arising from the exponential sensitivity of α_{xy} to the small gap $\Delta \approx 0.002v$. The Nernst response provides a far more sensitive probe of topological gap size than the Hall conductivity.

Figure 11(a-d) presents $\kappa^{(0)}_{xy}/T$ at $T = 0$ for all four phases. The top overlay panel and the four individual panels confirm the Wiedemann Franz relation $\kappa^{(0)}_{xy}/T|_{T=0} = (\pi^2/3)\sigma_{xy}$ throughout.

In the QSH phase, panel 11(a), $\kappa^{(0)}_{xy}/T$ is nearly zero in the gap and shows a small positive peak of $+0.080\, k_B^2/h$ near the band edge. The flatness in the gap is consistent with $\sigma_{xy} \approx 0$ there and the WF law.

In the $C = -1$, phase panel 11(b), $\kappa^{(0)}_{xy}/T$ develops a negative peak of $-0.025\, k_B^2/h$ at $E_F = 0$, tracking the negative σ_{xy} of this phase. By contrast, the $C = +1$ phase has a sharp positive peak of $+1.913\, k_B^2/h$, which is close to the quantized value $\pi^2 k_B^2/3h \approx \pi^2/3\, k_B^2/h$ expected for a $C = +1$ Chern insulator, panel 11(c), at $T \to 0$. The near quantization here is particularly significant: it demonstrates that the LPL induced Chern insulating phase carries a genuine topological thermal Hall response satisfying the anomalous WF law, placing it on the same place as magnetically ordered quantum anomalous Hall insulators.

The trivial phase, panel 11(d), shows a broad positive $\kappa^{(0)}_{xy}/T$ with peak $+1.023\, k_B^2/h$, again showing its non quantized σ_{xy} through the WF law. Comparing all, the most striking result is the sign structure: $\kappa^{(0)}_{xy}/T$ is positive for the QSH, $C = +1$, and trivial phases, but negative for the $C = -1$ phase. This sign change is a direct consequence of the sign of the Chern number and provides an unambiguous thermal Hall signature of the different topological phases.

The results reveal a clear hierarchy of sensitivity among the three transport coefficients. The AHC σ_{xy} is the most direct topological probe, approaching quantized values in the Chern phases and vanishing in the QSH gap. The ATHC $\kappa^{(0)}_{xy}/T$ mirrors σ_{xy} through the WF law, offering a thermal channel to the same topological information. The ANC α_{xy}, however, goes beyond this: it is exponentially sensitive to the gap size and provides a quantitative measure of how close the system is to

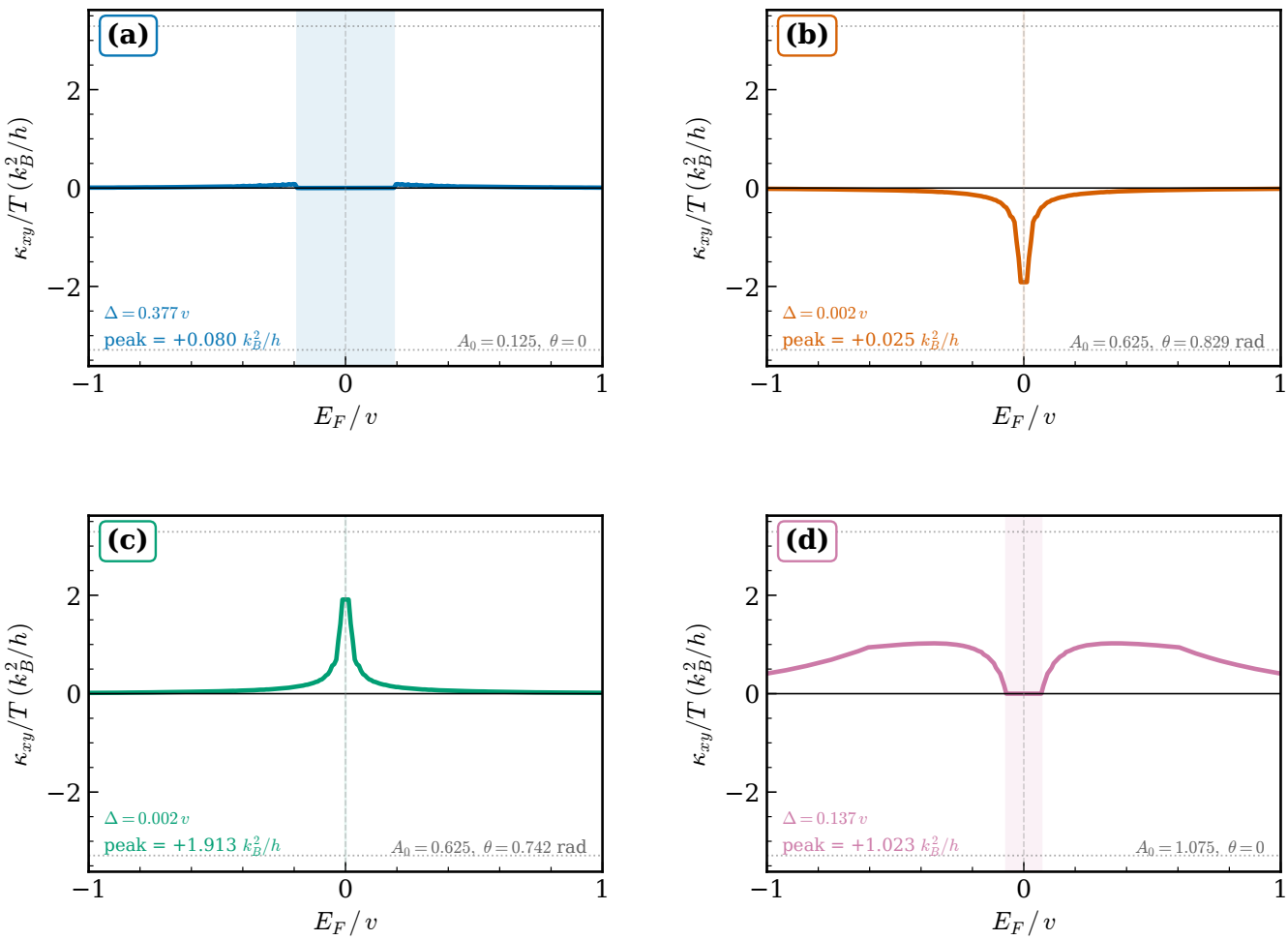

FIG. 11. Anomalous thermal Hall conductivity $\kappa_{xy}^{(0)}/T$ at $T = 0$ across all four topological phases. Top panel: overlay of all four phases; dotted lines at $\pm\pi^2/3\,k_B^2/h$ mark the WF-predicted quantized values. Individual panels (a)–(d) use the same parameters as Fig. 9. The $C = +1$ Chern phase (c) shows near-quantization $\kappa_{xy}^{(0)}/T \approx +1.913\,k_B^2/h \approx \pi^2/3\,k_B^2/h$, confirming the anomalous Wiedemann–Franz law $\kappa_{xy}^{(0)} = -L_0 T\sigma_{xy}$. The sign of $\kappa_{xy}^{(0)}/T$ tracks the sign of C, providing an unambiguous thermal Hall fingerprint of each topological phase.

a phase boundary. The enhancement of α_{xy} in the near critical Chern phases compared to the QSH phase makes the Nernst effect a natural experimental tool for detecting topological phase transitions driven by LPL intensity or polarization angle.

From an experimental perspective, the four transport signatures described here are all accessible in a single device geometry. The Hall conductivity σ_{xy} can be measured by transverse voltage in a Hall bar geometry under LPL irradiation. The Nernst coefficient α_{xy} requires an additional temperature gradient, achievable through laser heating of one contact. The thermal Hall conductivity $\kappa_{xy}^{(0)}$ is accessed by the transverse temperature difference under longitudinal heat current. The fact that all three quantities share the same polarization angle dependence vanishing at $\theta = \pm\pi/4$ and reversing sign under $\theta \to \pi/2 - \theta$ means that a single angle rotation measurement can confirm the altermagnetic origin of all observed responses simultaneously.

Table I summarizes the contrasting behavior of the AM and AFM under LPL irradiation across all transport and topological observables. The central distinction is that LPL breaks $C_{4z}\mathcal{T}$ in the AM but cannot break $\mathcal{PT}$ in the AFM. This propagates to every observable: the AFM has $\sigma_{xy} = \alpha_{xy} = \kappa_{xy}^{(0)} = \sigma_{xy}^s = \alpha_{xy}^s = \kappa_{xy}^{(0),s} = 0$ for all A_0, θ, and T. Only the QSH→Trivial transition (direct, no intermediate Chern phase) is possible in the AFM.

VI. MEASUREMENT OF THE ANOMALOUS TRANSPORT COEFFICIENTS IN EXPERIMENTS

Altermagnetic order, and configurable band topology, the essential components needed to implement our proposal, are currently present in a number of material

TABLE I. Comparison of transport properties and topological phase transitions in the d-wave AM and conventional AFM under LPL irradiation ($\theta = 0$, QSH initial phase).

Observable	AM ($t_a = \sqrt{3}$)	AFM ($t_a = 1$)
Symmetry under LPL	$C_{4z}\mathcal{T}$ broken	$\mathcal{PT}$ preserved
Spin degeneracy	Lifted (except $\theta = \pm\pi/4$)	Preserved everywhere
Phase sequence with A_0	QSH $\rightarrow$ Chern ($C = -1$) $\rightarrow$ Trivial	QSH $\rightarrow$ Trivial
A_0^{c1} (first transition)	≈ 0.521	≈ 0.640
A_0^{c2} (second transition)	≈ 0.918	—
σ_{xy}	Nonzero, light-controlled	Zero
α_{xy}	Nonzero, thermally activated	Zero
$\kappa_{xy}^{(0)}$	Nonzero; quantized in Chern phase	Zero
WF law $\kappa_{xy}^{(0)} = -L_0 T \sigma_{xy}$	Satisfied at $T \rightarrow 0$	Trivially ($0 = 0$)
Mott formula	Satisfied at low T	Trivially
Sign control by θ	All three coefficients	None
Nodes at $\theta = \pm\pi/4$	All three coefficients	All zero regardless
QSH spin transport	$\sigma_{xy}^s, \alpha_{xy}^s, \kappa_{xy}^{(0),s} \neq 0$	$\sigma_{xy}^s, \alpha_{xy}^s, \kappa_{xy}^{(0),s} \neq 0$

substrates. Recently discovered altermagnets like RuO_2 and MnTe-based compounds, as well as constructed heterostructures with customisable symmetry and band structure, are examples of candidate systems.

The off-resonant (high-frequency) approximation is valid because mid-infrared to terahertz radiation can be used to enter the Floquet driving regime considered in this work, where the photon energy exceeds the electronic bandwidth. Driving frequencies between 10 and 100 THz are adequate for typical hopping amplitudes on the order of 0.1 to 1 eV. The required field strengths correspond to dimensionless amplitudes $A_0 \sim 0.5$–1 , which are achievable with currently available pulsed laser systems. Crucially, the anticipated topological transitions rely on the field amplitude to frequency ratio, which permits experimental tuning flexibility without the need for extreme circumstances. Floquet-engineered phases can endure across picosecond timeframes, which is adequate for transport measurements, according to recent pump-probe experiments.

Standard measurement geometries allow for experimental access to the predicted transport signatures. Transverse voltage measurements under optical irradiation in a Hall bar design can be used to identify the anomalous Hall conductivity. By creating a longitudinal temperature gradient achievable by integrated microheaters or local laser heating and measuring the transverse voltage that results, one can quantify the anomalous Nernst effect. Using well-established thermometry methods, the thermal Hall conductivity can be derived from transverse temperature gradients. Notably, all three reactions exhibit a distinctive dependency on the input light's polarization angle, reversing sign under orthogonal rotation and disappearing at symmetry-restoring directions. In contrast to traditional antiferromagnets, this angular dependence offers a strong and clear indication of altermagnetic order.

Lastly, the sign reversal of the Hall responses by polarization rotation and the tunability of the Nernst peak temperature provide useful paths for device applications. For ultrafast spin caloritronics, it is very appealing to be able to control heat and charge transport solely optically, without the use of magnetic fields. The qualitative behaviors described here are symmetry-driven and so anticipated to be widely applicable across a broad class of altermagnetic systems, while quantitative predictions will require material-specific modeling.

VII. CONCLUSION

We have presented a comprehensive theoretical study of intrinsic thermal transport in a d-wave altermagnetics under linearly polarized light (LPL). We find that the Berry curvature can be expressed in closed form in both spin sectors, and that all anomalous transport responses in the altermagnetic (AM) phase have a microscopic origin due to the differential renormalization $j_1 \neq j_2$.

A main result is the emergence of a sequential topological phase transition, QSH $\rightarrow$ spin-polarized Chern insulator ($C = -1$) at $A_0^{c1} \approx 0.521$, followed by Chern $\rightarrow$ trivial at $A_0^{c2} \approx 0.918$. This hierarchy is unique to altermagnets, in contrast to conventional antiferromagnets (AFMs), which exhibit a direct QSH $\rightarrow$ trivial transition.

Within the Chern insulating regime, both charge and thermal responses become topologically quantized, with $\sigma_{xy} = -e^2/h$ and $\kappa_{xy}/T = \pi^2 k_B^2/(3h)$, and satisfy the anomalous Wiedemann–Franz relation $\kappa_{xy} = -L_0 T \sigma_{xy}$. In contrast, the anomalous Nernst conductivity α_{xy} is

purely thermally driven, vanishing at $T = 0$ and peaking at a characteristic temperature T^* tunable by A_0 and θ, in agreement with the Mott relation $\alpha_{xy} \approx -(\pi^2 k_B^2 T/3e)\,\partial_\mu \sigma_{xy}$.

All anomalous transport coefficients exhibit a universal d-wave dependence on the polarization angle, vanishing at $\theta = \pm\pi/4$, reaching extrema at $\theta = 0, \pm\pi/2$, and reversing sign under $\theta \to \pi/2 - \theta$. This provides a purely optical route to control and reverse the thermal Hall response, with no counterpart in conventional AFMs.

Crucially, thermal Hall and Nernst measurements are established as definitive experimental probes of altermagnetic order, even in regimes where charge transport signals may be misleading, due to the total absence of anomalous transport in AFMs under LPL, in sharp contrast to AMs.

With obvious implications for spin caloritronics and the identification of altermagnetic phases in developing quantum materials, these findings establish Floquet engineering of altermagnetic topological insulators as a potential platform for regulating thermal and spin transport.

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